Constructing Diversification Constraints

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1 Overview

Consider the function, $f : \mathbf{R}^n \to \mathbf{R}$, defined by[1]

$$f(\mathbf{x}) = \sum_{i=1}^{k} x_{[i]}.$$

That is, $f(\mathbf{x})$ is the sum of the k largest entries of $\mathbf{x}$. We are interested in optimization problems involving a vector $\mathbf{x}$ where the sum of the top k entries is bounded; that is, $f(\mathbf{x}) \leq M$ for a given M. How could we do this? One way is to write down all possible combinations of k elements from $\mathbf{x}$ and write a constraint that bounds their sum to be less than or equal to M. But the number of constraints that one has to write is $\binom{n}{k}$. This becomes combinatorially large very quickly. In the next sections we use LP duality to replace these $\binom{n}{k}$ constraints with $O(n)$ linear constraints.[2]

In finance, this constraint arises when one wants to create (or modify) a portfolio so that the new portfolio has the property that its top k securities (notionally) do not exceed some fixed percentage of its total notional.

[1]For a given vector, $\mathbf{x} \in \mathbf{R}^n$, create a vector $\mathbf{x}'$ by sorting $\mathbf{x}$ from highest to lowest (ties broken arbitrarily, as the sums below are independent of the tie-breaking rule). Define the notation, $x_{[i]}$ by: $x_{[i]} \equiv x'_i$. Note: For the rest of this paper we assume that $k \in \mathbf{Z}^+$ with $k \leq n$.

[2]A bold font is used to denote vectors. We refer to a vector of 0s as $\mathbf{0}$, and likewise we refer to a vector of 1s as $\mathbf{1}$. The mathematical expression, $\mathbf{a} \preceq \mathbf{b}$, is to be interpreted as the statement that every element of the vector $\mathbf{a}$ is less than or equal to the corresponding element in the vector $\mathbf{b}$. Similarly, the expression, $\mathbf{a} \succeq \mathbf{b}$, is the statement that each element of $\mathbf{a}$ is greater than or equal to each corresponding element of $\mathbf{b}$.

2 The Function f as an Optimization Problem

We wish to characterize $f(\mathbf{x})$ as the value of an optimization. We claim that for a given $\mathbf{x}$, $f(\mathbf{x})$ equals the optimal value of the following constrained *discrete* optimization problem:

$$\max_{\mathbf{y}\in\mathbf{Z}_2^n} \quad \mathbf{y}^T\mathbf{x} \tag{1}$$

$$\text{s.t.} \quad \mathbf{y}^T\mathbf{1} = k \tag{2}$$

This is $f(\mathbf{x})$ written as an LP. Although this optimization is succinct, there are still $\binom{n}{k}$ candidates to examine to find the optimal solution in this discrete setting.

Claim. The discrete optimization problem is equivalent to the following continuous problem:[3]

$$\begin{aligned} \max_{\mathbf{y}\in\mathbf{R}^n} \quad & \mathbf{y}^T\mathbf{x} \\ \text{s.t.} \quad & \mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} && (3) \\ & \mathbf{y}^T\mathbf{1} = k && (4) \end{aligned}$$

Proof. Since its feasible set contains the discrete problem's feasible set (every $\mathbf{y} \in \mathbf{Z}_2^n$ with $\mathbf{y}^T\mathbf{1} = k$ also satisfies $\mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1}$), the optimal value of the continuous problem is at least as large as the optimal value of the original discrete problem. We now show the reverse. For any optimal solution $\mathbf{y}^*$ of the continuous problem, we construct a $\mathbf{y}' \in \mathbf{Z}_2^n$ satisfying (2) and yielding the same objective value (1) as $\mathbf{y}^*$.

If all components of $\mathbf{y}^*$ are integers, by (3) they are each 0 or 1; in this case take $\mathbf{y}' = \mathbf{y}^*$.

Otherwise, consider the components of $\mathbf{y}^*$ that are non-integer; that is, have values strictly between 0 and 1. By (4), $\sum_i y_i^* = k$ is an integer; the integer-valued components contribute an integer to this sum, so the non-integer components must sum to an integer too. Consequently, there must be more than one such component. It must also be the case that amongst these non-integer components the corresponding "x" values are the same. Otherwise, if two of these components had differing "x" values, then the "y" component with the smaller "x" value could shift some of its value to the "y" component with the larger "x" value in such a way as to preserve the constraints, (3) and (4). But then we would have a larger objective value than the supposed maximum – contradiction. Therefore, all of the "x"

[3] The objective $\mathbf{y}^T\mathbf{x}$ is continuous and the feasible set is closed and bounded (a compact polytope in $\mathbf{R}^n$); the supremum is therefore attained, so we may write "max" instead of "sup".

components associated with non-integer "y" values must have the same value. As previously stated, the sum of these non-integer values is an integer, which is necessarily less than the number of non-integer components in $\mathbf{y}^*$. Let j denote this integer sum. Pick any j of these non-integer values, and let J be the set of their index values in $\mathbf{y}^*$. Let $\tilde{J}$ be the set of indices corresponding to the remaining non-integer values of $\mathbf{y}^*$.

Now, create a new vector $\mathbf{y}' \in \mathbf{Z}_2^n$ and set its values in the following way: Set the values of $\mathbf{y}'$ corresponding to the indices in J to be 1. Set the values of $\mathbf{y}'$ corresponding to the indices in $\tilde{J}$ to 0. Set the rest of the components of $\mathbf{y}'$ to the corresponding values from $\mathbf{y}^*$. By our construction, ${\mathbf{y}'}^T\mathbf{1} = k$. The components of $\mathbf{y}'$ inherited from $\mathbf{y}^*$ were already integers, and by (3) every integer component of $\mathbf{y}^*$ is 0 or 1. It follows that $\mathbf{y}' \in \mathbf{Z}_2^n$ and satisfies the constraint of the discrete problem, (2). The objective value is also preserved: let x_c denote the common value of the "x" components on $J \cup \tilde{J}$. The contribution to the objective from those indices equals $x_c \sum_{i \in J \cup \tilde{J}} y_i^* = x_c\, j$, which is exactly the contribution from $\mathbf{y}'$ on the same indices. Therefore, $\mathbf{y}'$ is in the feasible set of the discrete problem and attains the optimal value of the continuous problem; that is, the optimal value of the discrete problem is at least the optimal value of the continuous problem. Consequently, the two optimization problems are equivalent.

The continuous problem above is equivalent to the following minimization:

$$\min_{\mathbf{y} \in \mathbf{R}^n} \quad -\mathbf{y}^T\mathbf{x} \tag{5}$$

$$\text{s.t.} \quad \mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} \tag{6}$$

$$\mathbf{y}^T\mathbf{1} = k \tag{7}$$

Treating $\mathbf{x}$ as a constant, this is a *convex* optimization problem.

3 A Dual Description of the Optimization

Since the problem described by (5)–(7) is *convex*, the value of its solution is the same as the value of its associated *dual* problem.[4]

[4]Normally one needs to show that a convex problem satisfies Slater's condition in order to invoke strong duality. For linear programs, strong duality holds whenever the primal is feasible and bounded – no Slater check needed. The bilinear term $-\mathbf{y}^T\mathbf{x}$ becomes linear in $\mathbf{y}$ once we fix $\mathbf{x}$ as a parameter.

To form the dual problem we need the Lagrangian, which is:[5]

$$L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = -\mathbf{y}^T\mathbf{x} - \boldsymbol{\lambda}_1^T\mathbf{y} + \boldsymbol{\lambda}_2^T(\mathbf{y} - \mathbf{1}) + \nu(\mathbf{y}^T\mathbf{1} - k).$$

Here $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2 \in \mathbf{R}^n$ and $\nu \in \mathbf{R}$ are the *dual* variables.

Define g by:

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} \; L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}).$$

Substituting for L this becomes

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} \Big[\mathbf{y}^T \big(-\mathbf{x} - \boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 + \nu \mathbf{1} \big) - \mathbf{1}^T \boldsymbol{\lambda}_2 - \nu\, k \Big].$$

The dual problem is then

$$\max_{\substack{\boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \nu}} \quad g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}).$$

Which is[6]

$$\begin{aligned} \max_{\substack{\boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \nu}} \quad & -\mathbf{1}^T \boldsymbol{\lambda}_2 - \nu\, k \\ \text{s.t.} \quad & -\boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 + \nu \mathbf{1} = \mathbf{x} \end{aligned}$$

This is equivalent to[7]

$$\begin{aligned} \max_{\substack{\boldsymbol{\lambda} \succeq \mathbf{0} \\ \nu}} \quad & -\nu\, k - \mathbf{1}^T \boldsymbol{\lambda} \\ \text{s.t.} \quad & \mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \end{aligned}$$

[5] $\mathbf{x}$ is a parameter (constant for the optimization) – separated from the optimization variables by a semi-colon.

[6] Note that g is $-\infty$ unless the coefficient of $\mathbf{y}$ is the zero vector. Consequently, the maximum must necessarily occur where that coefficient is zero.

[7] We can remove n variables and the n equality constraints by eliminating $\boldsymbol{\lambda}_1$. The equality $-\boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 + \nu \mathbf{1} = \mathbf{x}$ together with $\boldsymbol{\lambda}_1 \succeq \mathbf{0}$ is equivalent to $\boldsymbol{\lambda}_2 + \nu \mathbf{1} \succeq \mathbf{x}$. Since there is now only one $\boldsymbol{\lambda}$, we relabel $\boldsymbol{\lambda}_2$ as $\boldsymbol{\lambda}$.

Negating the objective converts the max to a min:

$$\begin{aligned} \min_{\boldsymbol{\lambda},\nu} \quad & \nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \\ \text{s.t.} \quad & \mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \\ & \boldsymbol{\lambda} \succeq \mathbf{0} \end{aligned}$$

4 Characterization of f as $O(n)$ Linear Constraints

We now apply the results of the previous sections to the following problem: Add constraints to an optimization problem that includes a *variable* $\mathbf{x} \in \mathbf{R}^n$ so that $f(\mathbf{x}) \leq M$ for a given M.

We claim that one can bound $f(\mathbf{x})$ by adding $n+1$ new scalar variables (a vector $\boldsymbol{\lambda} \in \mathbf{R}^n$ and a scalar $\nu \in \mathbf{R}$), along with the following:

$$\nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \leq M \tag{8}$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \tag{9}$$

$$\boldsymbol{\lambda} \succeq \mathbf{0} \tag{10}$$

Why? Because for any $(\boldsymbol{\lambda}, \nu)$ satisfying (9) and (10), $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ is an upper bound on $f(\mathbf{x})$ – this is weak duality. Consequently, we have the inequality: $f(\mathbf{x}) \leq \nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \leq M$. The only concern remaining is that there may be a gap between the values of $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ and $f(\mathbf{x})$ – making (8) too restrictive. But this is not the case: by strong duality, the minimum of $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ over $\boldsymbol{\lambda}, \nu$ (subject to (9) and (10)) equals $f(\mathbf{x})$. So there always exist $\boldsymbol{\lambda}, \nu$ feasible for (9) and (10) with $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda} = f(\mathbf{x})$, and (8) is satisfiable for $\mathbf{x}$ exactly when $f(\mathbf{x}) \leq M$ – no extra restriction on $\mathbf{x}$ is introduced.

One might ask why we couldn't do exactly the same procedure with the "simpler" characterization we had with equations (5)–(7), rather than going through so much effort with a dual problem. It would require the same number of variables and constraints.

We could try that; however, note that once we take the characterization of a bound on a "given" $\mathbf{x}$ and place it into an optimization problem, that "given" $\mathbf{x}$ is no longer a constant. It is now part of the optimization. The primal characterizes $f(\mathbf{x})$ as a *maximum* over $\mathbf{y}$; to enforce $f(\mathbf{x}) \leq M$ we would need $\mathbf{y}^T \mathbf{x} \leq M$ for *every* feasible $\mathbf{y}$ – which is exactly the original combinatorial blow-up. The only way to collapse it into a finite set of constraints is to encode the LP's optimality conditions, which introduce bilinear terms in the (now) *variable* $\mathbf{x}$ and the new *variable* $\mathbf{y}$. These terms are decidedly *non-linear*; in fact, they are not even *convex*.

In contrast to this "simpler" characterization and the original naïve combinatorial approach, we have shown that by examining an associated dual problem, $f(\mathbf{x})$ can be bounded by adding $(n + 1)$ new variables and $(2n + 1)$ *linear* constraints.